Assignment 4

14X062 Computational Imaging

Recently, the production of three dimensional (3D) images obtained by X-ray computed tomography (CT) experiments is gaining increasing interest by the scientific community. This is due to the big potential of this technique for the nondestructive description of the internal microstructure of opaque objects.

To investigate particularities of parallel-beam CT system simulation based on the Algoton package (https://github.com/algotom/algotom) can be used which contains the following processing pipeline

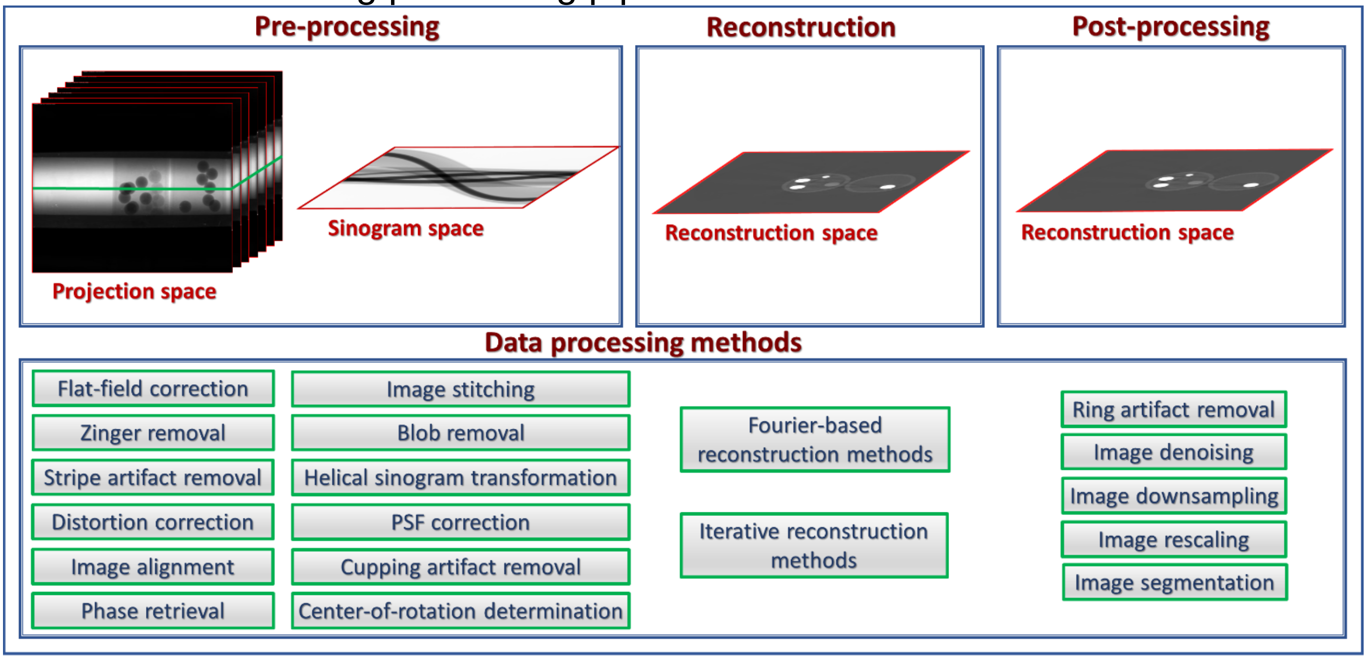

Fig. 1. Steps for 3D/slice reconstruction in X-ray tomography.

The detailed description of Algoton package is given here https://algotom.readthedocs.io

Task 1. Install Algoton package (see https://algotom.readthedocs.io/en/latest/toc/section3.html) and take into account the following:

- If users install Algotom to an existing environment and Numba fails to install due to the latest Numpy:
  - Downgrade Numpy and install Algotom/Numba again.
  - Create a new environment and install Algotom first, then other packages.
  - Use conda instead of pip.
- Avoid using the latest version of Python or Numpy as the Python ecosystem taking time to keep up with these twos.

Task 2. (for familiarization) Perform few slides reconstruction (see https://algotom.readthedocs.io/en/latest/toc/section4/section4_5.html#reconstructing-several-slices)

Task 3. (for familiarization) Compare different reconstruction methods (https://algotom.readthedocs.io/en/latest/toc/section4/section4_5.html#choosing-a-reconstruction-method)

- DFI method
- FBP method
- gridrec method

Task 4. (for familiarization) Extend the list of reconstruction methods with SART method from ASTRA Toolbox (https://astra-toolbox.com/docs/algs/SART.html)

Task 5. Compare different methods of stripe/ring removal (https://algotom.readthedocs.io/en/latest/toc/section4/section4_4.html#all-types-of-ring-artifacts):

- wavelet-fft-based method
- fft-based method:
- normalization-based method:
- regularization-based method:
- sorting-based method

using the data from (https://github.com/nghia-vo/sarepy/tree/master/data)

To complete all results (report together with your code) into archive and submit it to Computational imaging course moodle.